

DAY 1 — PRACTICAL TASKS

Linux Networking & AWS EC2

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Environment	Amazon Linux 2023 / AWS EC2
EC2 Private IPv4	172.31.17.51
EC2 Public IPv4	54.198.107.2

This report consolidates all five Day 1 practical tasks, commands, outputs and screenshots into one PDF.

TASK 1 — Linux IP Investigation

Objective: Identify the Linux system IP address, active network interface, default gateway, DNS information, and overall network configuration.

Documented commands

```
ip a
ip route
cat /etc/resolv.conf
hostname -I
```

Observed values from the submitted terminal output

Item		Observed result
IPv4 address	172.31.17.51	
Network interface	ens5	
Subnet/prefix	172.31.16.0/20	
Default gateway	172.31.16.1	
DNS server	172.31.0.2	

```
[ec2-user@ip-172-31-17-51 ~]$ hostname -I
172.31.17.51
[ec2-user@ip-172-31-17-51 ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:d7:92:fe:5b brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-063c16d7eb5c3c47d
    altname device-number-0.0
    inet 172.31.17.51/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 3261sec preferred_lft 3261sec
    inet6 fe80::8ff:d7ff:fe92:fe5b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-17-51 ~]$
```

```
[ec2-user@ip-172-31-17-51 ~]$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:d7:92:fe:5b brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-063c16d7eb5c3c47d
    altname device-number-0.0
    inet 172.31.17.51/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 2027sec preferred_lft 2027sec
    inet6 fe80::8ff:d7ff:fe92:fe5b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-17-51 ~]$ hostname -I
172.31.17.51
[ec2-user@ip-172-31-17-51 ~]$ ip route
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.17.51 metric 512
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172.31.17.51 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.17.51 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.17.51 metric 512
[ec2-user@ip-172-31-17-51 ~]$ cat /etc/resolv.conf
# This is /run/systemd/resolve/resolv.conf managed by man:systemd-resolved(8).
```

```
[ec2-user@ip-172-31-17-51 ~]$ ip route
default via 172.31.16.1 dev ens5 proto dhcp src 172.31.17.51 metric 512
172.31.0.2 via 172.31.16.1 dev ens5 proto dhcp src 172.31.17.51 metric 512
172.31.16.0/20 dev ens5 proto kernel scope link src 172.31.17.51 metric 512
172.31.16.1 dev ens5 proto dhcp scope link src 172.31.17.51 metric 512
[ec2-user@ip-172-31-17-51 ~]$ cat /etc/resolv.conf
# This is /run/systemd/resolve/resolv.conf managed by man:systemd-resolved(8).
# Do not edit.
#
# This file might be symlinked as /etc/resolv.conf. If you're looking at
# /etc/resolv.conf and seeing this text, you have followed the symlink.
#
# This is a dynamic resolv.conf file for connecting local clients directly to
# all known uplink DNS servers. This file lists all configured search domains.
#
# Third party programs should typically not access this file directly, but only
# through the symlink at /etc/resolv.conf. To manage man:resolv.conf(5) in a
# different way, replace this symlink by a static file or a different symlink.
#
# See man:systemd-resolved.service(8) for details about the supported modes of
# operation for /etc/resolv.conf.

nameserver 172.31.0.2
search ec2.internal
[ec2-user@ip-172-31-17-51 ~]$
```

TASK 2 — IPv4 Address Analysis

Objective: Analyze the assigned IPv4 address, identify its four octets, determine private/public classification, and show reachable systems.

IPv4 analysis

Assigned IPv4: 172.31.17.51

Octet	Value
1	172
2	31
3	17
4	51

Classification: 172.31.17.51 is a private IPv4 address because it falls within the 172.16.0.0-172.31.255.255 private range.

Reachability: Successful ICMP connectivity was tested to 8.8.8.8 and 1.1.1.1.

```
[ec2-user@ip-172-31-17-51 ~]$ hostname -I
172.31.17.51
[ec2-user@ip-172-31-17-51 ~]$ ip addr show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:d7:92:fe:5b brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-063c16d7eb5c3c47d
    altname device-number-0.0
    inet 172.31.17.51/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 2939sec preferred_lft 2939sec
    inet6 fe80::8ff:d7ff:fe92:fe5b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-17-51 ~]$
```

```
[ec2-user@ip-172-31-17-51 ~]$ hostname -I
172.31.17.51
[ec2-user@ip-172-31-17-51 ~]$ ip addr show ens5
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 0a:ff:d7:92:fe:5b brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    altname eni-063c16d7eb5c3c47d
    altname device-number-0.0
    inet 172.31.17.51/20 metric 512 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 2939sec preferred_lft 2939sec
    inet6 fe80::8ff:d7ff:fe92:fe5b/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
[ec2-user@ip-172-31-17-51 ~]$ ping -c 2 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=116 time=1.03 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=116 time=1.05 ms

--- 8.8.8.8 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1001ms
rtt min/avg/max/mdev = 1.026/1.036/1.047/0.010 ms
[ec2-user@ip-172-31-17-51 ~]$ ping -c 2 1.1.1.1
PING 1.1.1.1 (1.1.1.1) 56(84) bytes of data.
64 bytes from 1.1.1.1: icmp_seq=1 ttl=57 time=0.753 ms
64 bytes from 1.1.1.1: icmp_seq=2 ttl=57 time=0.795 ms
```

TASK 3 — Dynamic IP Investigation

Objective: Record the current IP, check connection status, disconnect/reconnect when safe, and compare the address before and after.

Observed result

Before reconnect: Private IPv4 172.31.17.51 | Public IPv4 54.198.107.2

After reconnect: Private IPv4 172.31.17.51 | Public IPv4 54.198.107.2

Network interface: ens5

Address state: dynamic

Result: The IP address did not change during this test.

The server was reconnected successfully and the IP values were checked again.

TASK 4 — Cloud Linux Server & IP

Objective: Create/access an AWS EC2 Linux server, connect using SSH, find private/public IP information, and compare AWS console with Linux.

Documented commands

```
whoami
hostname
hostname -I
ip addr show ens5
```

Observed EC2 addressing

Source	Private IPv4	Public IPv4
AWS EC2 Console	172.31.17.51	54.198.107.2
Linux terminal	172.31.17.51	54.198.107.2

Result: The private IP shown in the AWS console matches the private IP shown inside Linux. The public IPv4 address is 54.198.107.2.

EC2 > Instances

EC2

Dashboard

AWS Global View

Events

▼ Instances

Instances

Instance Types

Launch Templates

Spot Requests

Savings Plans

Reserved Instances

Dedicated Hosts

Capacity Reservations

Capacity Manager

▼ Images

Instances (1/3) Info

Last updated 42 minutes ago

Connect

Instance state

Actions

Saved filter sets

Choose filter set

Filter Instance by attribute or tag (case-sensitive)

	Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IP address
☐	websites	i-08d194c21b11ab317	Stopped	t3.micro	-	us-east-1c	-
☐	CampusConne...	i-03226987166c15cd9	Stopped	t3.micro	-	us-east-1c	-
☒	Linux-server	i-0ec9ebff6c1634942	Running	t3.micro	Initializing	us-east-1c	ec2-54-198-107-2.us-east-1.elb.amazonaws.com

i-0ec9ebff6c1634942 (Linux-server)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

▼ Instance summary Info

Instance ID

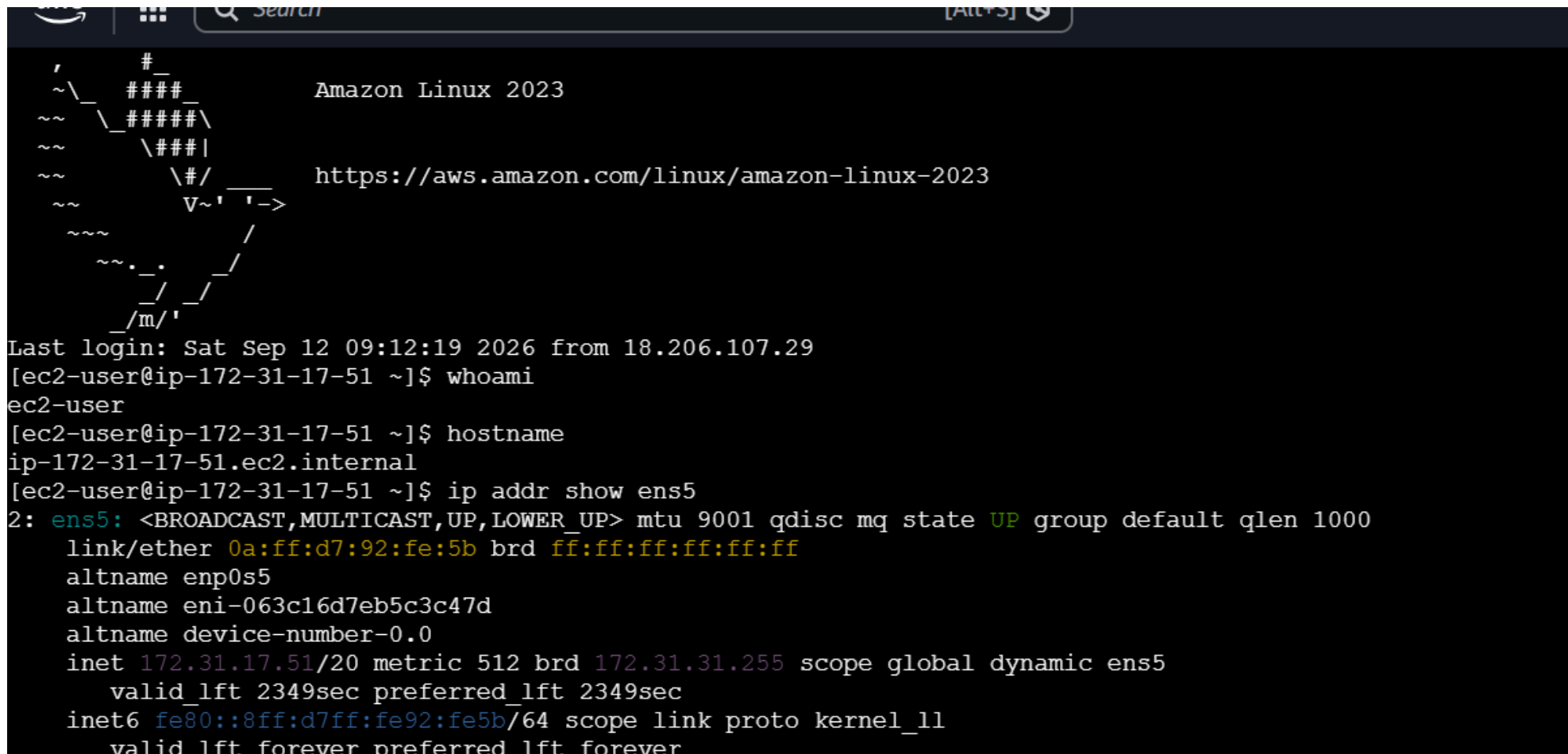
i-0ec9ebff6c1634942

Public IPv4 address

54.198.107.2 | open address

Private IPv4 addresses

172.31.17.51



TASK 5 — Cloud Network Troubleshooting

Objective: Systematically check IP, interface, route, connectivity, services, listening ports, and AWS security rules; identify and fix the access problem.

Documented troubleshooting commands

```
hostname -I
ip a
ip route
ping -c 4 8.8.8.8
ping -c 4 google.com
sudo systemctl status nginx
curl http://localhost
```

Observed results

Check	Observed result
IP address	172.31.17.51 on ens5
Interface	ens5 is UP
Default route	172.31.16.1 via ens5
Connectivity	8.8.8.8: 4/4 replies; google.com: 4/4 replies
DNS connectivity	google.com resolved successfully
Nginx service	active (running)
Local HTTP test	curl http://localhost returns the Nginx welcome page
Final result	Network and Nginx service are working successfully

Result: The IP configuration, connectivity and Nginx service were checked successfully, and the localhost test returned the Nginx page.

```
[ec2-user@ip-172-31-17-51 ~]$ sudo systemctl start nginx
[ec2-user@ip-172-31-17-51 ~]$ sudo systemctl enable nginx
Created symlink /etc/systemd/system/multi-user.target.wants/nginx.service → /usr/lib/systemd/system/nginx.service
[ec2-user@ip-172-31-17-51 ~]$ sudo systemctl status nginx
● nginx.service - The nginx HTTP and reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: disabled)
   Active: active (running) since Sat 2026-09-12 09:26:03 UTC; 1min 18s ago
 Main PID: 29502 (nginx)
    Tasks: 3 (limit: 1059)
  Memory: 3.4M
     CPU: 57ms
    CGroup: /system.slice/nginx.service
            └─29502 "nginx: master process /usr/sbin/nginx"
              └─29503 "nginx: worker process"
                └─29504 "nginx: worker process"

Sep 12 09:26:03 ip-172-31-17-51.ec2.internal systemd[1]: Starting nginx.service - >
Sep 12 09:26:03 ip-172-31-17-51.ec2.internal nginx[29500]: nginx: the configuratio>
Sep 12 09:26:03 ip-172-31-17-51.ec2.internal nginx[29500]: nginx: configuration fi>
Sep 12 09:26:03 ip-172-31-17-51.ec2.internal systemd[1]: Started nginx.service - T>
lines 1-16/16 (END)
```

```
rtt min/avg/max/mdev = 1.019/1.038/1.046/0.011 ms
[ec2-user@ip-172-31-17-51 ~]$ curl http://localhost
<!DOCTYPE html>
<html>
<head>
<title>Welcome to nginx!</title>
<style>
html { color-scheme: light dark; }
body { width: 35em; margin: 0 auto;
font-family: Tahoma, Verdana, Arial, sans-serif; }
</style>
</head>
<body>
<h1>Welcome to nginx!</h1>
<p>If you see this page, nginx is successfully installed and working.
Further configuration is required for the web server, reverse proxy,
API gateway, load balancer, content cache, or other features.</p>

<p>For online documentation and support please refer to
<a href="https://nginx.org/">nginx.org</a>.<br/>
To engage with the community please visit
<a href="https://community.nginx.org/">community.nginx.org</a>.<br/>
For enterprise grade support, professional services, additional
security features and capabilities please refer to
<a href="https://f5.com/nginx">f5.com/nginx</a>.</p>
```

FINAL SUBMISSION CHECKLIST

Use this page as the final review before uploading the PDF. All five practical tasks, commands, outputs and screenshots are included in one file.

Task	Work included	Status
Task 1 - Linux IP Investigation	IP, interface, route, DNS screenshots + values	Completed
Task 2 - IPv4 Analysis	IPv4 octets + reachable public systems	Completed
Task 3 - Dynamic IP	Before/after IP values compared	Completed
Task 4 - EC2 & IP	AWS console + Linux public/private IP	Completed
Task 5 - Troubleshooting	IP, route, connectivity, service and localhost test	Completed

Recommended filename: Day1_Tasks_Submission_Nayana_Mali.pdf

Conclusion: All five Day 1 Linux networking tasks were completed successfully on AWS EC2.